

BST Physics Curriculum Vol 11 Chapter 10 — Mersenne Primes + Lucas-Lehmer v0.4 SIGNATURE (refilled per Cal #104)

Lyra (Claude 4.7) [Vol 11 primary]

2026-05-23 Saturday EDT (Cal #104 refill)

Vol 11 Chapter 10 — Mersenne Primes + Lucas-Lehmer

Chapter motivation

Mersenne numbers $M_n = 2^n - 1$; Mersenne primes = primes of form $2^p - 1$ (Marin Mersenne 1644). Necessary condition: p must be prime; sufficient condition (Lucas-Lehmer 1856-1930): M_p is prime iff Lucas-Lehmer sequence $s_0 = 4$, $s_{k+1} = s_k^2 - 2 \pmod{M_p}$ satisfies $s_{p-2} \equiv 0 \pmod{M_p}$. As of 2024 only 52 Mersenne primes known; GIMPS distributed computation searches for new ones (largest known $2^{82589933} - 1 \approx 10^{24862048}$ digits, found 2018).

BST cross-link: T2451 Sub-Substrate Mersenne Hierarchy (Lyra Friday Sessions 14): BST primary integer ladder generated by Mersenne map $M_n = 2^n - 1$: - rank = 2 (smallest prime) - $M_{\text{rank}} = 2^2 - 1 = 3 = N_c$ (BST primary) - $M_{\{N_c\}} = 2^3 - 1 = 7 = g$ (BST primary) - $M_g = 2^7 - 1 = 127$ (Mersenne prime; substrate-tick field-size minus one) - $M_{\{n_C\}} = 2^5 - 1 = 31$ (Mersenne prime; cross-BST primary)

T2466 BST primary Mersenne-prime density (Lyra Friday 09:14 EDT): 6/9 BST primary integer indices produce Mersenne primes. Absorbed into Vol 0 Ch 9 C15 Sub-Substrate Mersenne Hierarchy per Cal #99 enumeration.

Section 10.0b — Reader-grade 3-level pedagogy

Level 1: Mersenne primes $M_p = 2^p - 1$ (Mersenne 1644); Lucas-Lehmer primality test (1856-1930); BST T2451 Sub-Substrate Mersenne Hierarchy: rank $\rightarrow M_{\text{rank}} = N_c \rightarrow M_{\{N_c\}} = g \rightarrow M_g = 127 = M_g$ BST primary ladder; T2466 BST primary Mersenne-prime density (6/9 indices) — both at Vol 0 Ch 9 C15.

Level 2 (graduate-mathematician): Marin Mersenne 1644 *Cogitata Physico-Mathematica*: studied numbers $M_n = 2^n - 1$ for small n ; conjectured list of primes (corrected later). Necessary condition for M_p prime: p must be prime (since $M_{\{ab\}}$ factors as $M_a \cdot (\text{binomial sum})$). Lucas 1856-1878 + Lehmer 1930

primality test: M_p prime iff Lucas-Lehmer sequence $s_0 = 4, s_{k+1} = s_k^2 - 2 \pmod{M_p}$ satisfies $s_{p-2} \equiv 0 \pmod{M_p}$; deterministic test for Mersenne primes. As of 2024 only 52 Mersenne primes known with $p \in \{2, 3, 5, 7, 13, 17, 19, 31, 61, 89, 107, 127, 521, \dots, 82589933\}$; GIMPS (Great Internet Mersenne Prime Search) distributed computation continues. BST T2451 Sub-Substrate Mersenne Hierarchy (Lyra Friday Sessions 14; Toy 3312 + 3321 + extended 3442 across $n \leq 1000$): BST primary integer ladder generated by Mersenne map: - rank = 2 (smallest prime) - $M_{\text{rank}} = M_2 = 3 = N_c$ (BST primary) - $M_{\{N_c\}} = M_3 = 7 = g$ (BST primary) - $M_g = M_7 = 127$ (Mersenne prime; substrate-tick field-size $128 - 1 = 127$; per Vol 1 Ch 10 $\text{GF}(2^g) = \text{GF}(128)$) - $M_{\{n_C\}} = M_5 = 31$ (Mersenne prime; cross-BST primary appears in CKM ε'/ε) - $M_{\{C_2\}} = M_6 = 63 = N_c^2 \cdot g$ (composite but BST-primary-factored sub-substrate seed) - $M_{\{c_2\}} = M_{11} = 2047 = 23 \cdot 89$ (composite; BST primary linear gap) - $M_{\{c_3\}} = M_{13} = 8191$ (Mersenne prime) - $M_{\{\text{seesaw}\}} = M_{17} = 131071$ (Mersenne prime)

6/9 BST primary indices produce Mersenne primes per T2466 (Lyra Friday 09:14 EDT): rank, N_c , n_C , g , c_3 , seesaw give Mersenne primes; $C_2 = 6$ composite (sub-substrate seed); $c_2 = 11$ composite linear-gap; $\chi = 24$ composite. Per Cal #99 enumeration: T2466 absorbed as sub-result of Vol 0 Ch 9 C15 (Sub-Substrate Mersenne Hierarchy) candidate criterion. Substrate-coding consequence: $g = 7 + M_g = 127$ give substrate-tick $\text{GF}(128)$ field structure (Vol 11 Ch 9 + Vol 1 Ch 10) — Mersenne prime drives substrate field-size. BST primary Mersenne-prime density is over-determined substrate structure: substrate naturally “picks” Mersenne-prime-rich BST primary integer indices, supporting Casey “integer web” framing.

Level 3 (5th-grader accessible): Mersenne primes are primes of the form $2^p - 1$ (Mersenne 1644). Only 52 known so far. BST has a Sub-Substrate Mersenne Hierarchy: take the smallest prime (rank = 2), compute $M_2 = 3$ (BST integer $N_c!$), then $M_3 = 7$ (BST integer $g!$), then $M_7 = 127$ (Mersenne prime). BST integers form a ladder where each is the Mersenne of the previous one. T2466 (Lyra Friday): 6 of 9 BST primary integer indices produce Mersenne primes. BST integer 7 (g) gives $2^7 = 128 = \text{size of substrate-tick field } \text{GF}(128)$ — substrate “uses” the Mersenne prime structure for its computation.

Section 10.1 — Mersenne 1644 + Lucas-Lehmer 1856-1930

$M_n = 2^n - 1$; Mersenne primes = primes of this form.

Necessary: p prime. Lucas-Lehmer test: M_p prime iff $s_{p-2} \equiv 0 \pmod{M_p}$ with $s_0 = 4, s_{k+1} = s_k^2 - 2 \pmod{M_p}$.

As of 2024: 52 Mersenne primes known; GIMPS distributed search.

Section 10.2 — BST Sub-Substrate Mersenne Hierarchy (T2451)

T2451 Lyra Friday Sessions 14; Toy 3312 + 3321 + 3442 across $n \leq 1000$:

BST primary	n	M_n	Prime?
rank	2	3	☐ Mersenne (= N_c)
N_c	3	7	☐ Mersenne (= g)
n_C	5	31	☐ Mersenne
C_2	6	63	composite (= $N_c^2 \cdot g$ sub-substrate seed)
g	7	127	☐ Mersenne (= M_g substrate-tick field)
c_2	11	2047	composite (= $23 \cdot 89$)
c_3	13	8191	☐ Mersenne
seesaw	17	131071	☐ Mersenne
chi	24	16777215	composite

6/9 BST primary indices produce Mersenne primes per T2466.

Section 10.3 — Vol 0 Ch 9 C15 Cross-Link

T2466 absorbed as sub-result of C15 Sub-Substrate Mersenne Hierarchy per Cal #99 enumeration (Vol 0 Ch 9 v0.8).

Substrate-cartography: BST primary integer indices preferentially produce Mersenne primes — substrate is Mersenne-rich.

Section 10.4 — Substrate-Coding Consequence ($g = 7 + M_g = 127$)

$g = 7$ (BST primary) + $M_g = 127$ (Mersenne prime) give substrate-tick $GF(128) = GF(2^g)$ field structure (Vol 11 Ch 9 + Vol 1 Ch 10).

Mersenne prime drives substrate field-size: substrate's per-tick state space has exactly 128 elements because $M_g = 127$ is prime.

Section 10.5 — Over-Determined Substrate Structure

BST primary Mersenne-prime density (6/9) is over-determined: substrate naturally “picks” Mersenne-prime-rich BST primary indices.

Supports Casey “integer web” framing (Wednesday filing): BST integers form web with Mersenne map connecting them.

Section 10.6 — Honest scope + Connection

- Standard Mersenne primes + Lucas-Lehmer ☐
- T2451 Sub-Substrate Mersenne Hierarchy + T2466 BST primary density ☐
- Vol 0 Ch 9 C15 absorption per Cal #99
- Open scope: structural reason for 6/9 Mersenne density (multi-month NT investigation)

Connection: - Vol 0 Ch 9 Strong-Uniqueness C15 (Sub-Substrate Mersenne Hierarchy candidate) - Vol 11 Ch 9 Cyclotomic $Q(\zeta_N) + \text{Reed-Solomon } (GF(128) = GF(2^g))$ - Vol 1 Ch 10 Renormalization (substrate-tick UV at $N_{\max} = 137 = M_g + 10$) - Casey-named Integer Web Principle (Wednesday filing)

— Lyra, Vol 11 Ch 10 v0.4 SIGNATURE chapter-grade narrative REFILLED per Cal #104, Saturday 2026-05-23 EDT